



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



KALPVUKSH 2.0 MINI HACKATHON

- **Problem Statement ID** – P16
- **Problem Statement Title** –
**Exposure of Personal Identity
Through Routine Online Sign-
Ups**
- **Domain** – Cybersecurity
- **Team ID** – T049
- **Team Name** – Cryptic_Vruksh



CRYPTIC_VRUKSH

SECURE · ROOTED · RESILIENT

Project: **TraceMask**

A different you for every website —
and proof of who leaked it.

TRACEMASK: IDENTITY FIREWALL FOR ONLINE SIGN-UPS

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

TraceMask is a local-first web app that **gives every sign-up its own real, deliverable email alias**, checks each website live before any data is shared, **automatically proves which company leaked or sold your identity**, and turns that proof into a PDF evidence report and a DPDP/GDPR erasure request.

❖ Detailed explanation of the proposed solution



1 SCAN before sign-up

Live risk score (0–100) from TLS, domain age (RDAP), breach history (HIBP), trackers, privacy policy and DMARC.



2 MASK with one alias per site

Real, deliverable alias (you+site-x7k2p@...) with a trust tier T0 Burner, T1 Tracked, T2 Protected or T3 Real.



3 TRACE leaks automatically

IMAP IDLE reads only headers; any sender other than that site is a leak, proven by SPF/DKIM/DMARC.



4 ACT on the evidence

Auto-quarantine, SHA-256 evidence PDF and a one-click erasure letter under DPDP Act 2023 / GDPR.

❖ How it addresses the problem

P16 pain point	TraceMask answer
A real address is a persistent identifier	Unique alias per site, so companies have nothing to link across services
Users can't tell who leaked or sold the address	Automatic leak attribution with authenticated-sender proof
Unsubscribe stops mail but keeps the data	Evidence-backed erasure request with 30-day follow-up
Fake data breaks account recovery	Aliases are real: resets always arrive (T2 never auto-blocked)
Many identities are hard to manage	Created in one click and all tracked in one dashboard
Sites deserve different trust	Tier T0–T3 from live risk score + purpose of sign-up

❖ Innovation and uniqueness of the solution



Leak attribution, not just aliasing

Names the leaking company within seconds, with signed-sender proof.



Risk-aware identity

Live scan + purpose decide the tier: burner, tracked, protected or real.



Evidence-grade output

SHA-256 hashed headers, chain of custody, DPDP-referenced PDF.



Private by design

Runs on 127.0.0.1, zero third-party packages, encrypted vault.

Capability	Alias services*	TraceMask
Automatic leak attribution	Manual	Yes
Pre-sign-up risk scan	No	Yes
Evidence PDF + erasure letter	No	Yes

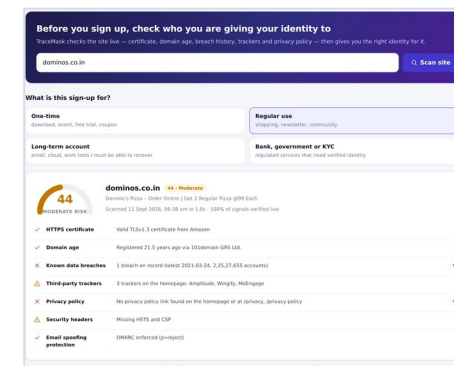
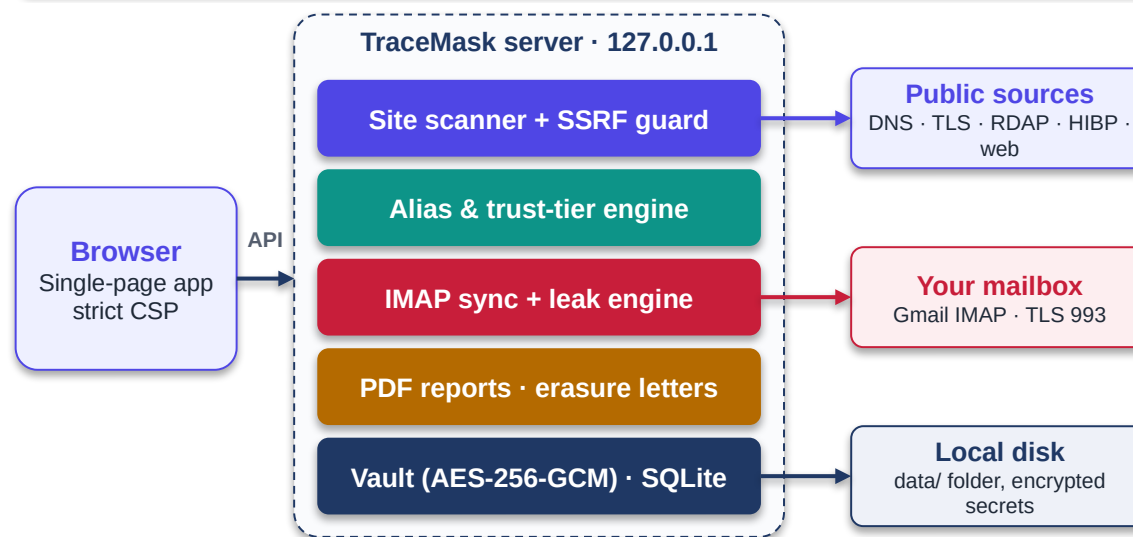
*Firefox Relay, SimpleLogin, DuckDuckGo, addy.io (see ref. 12)

TECHNICAL APPROACH

❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

-  **Node.js 24 (JavaScript)**
Zero third-party npm packages
-  **node:sqlite**
Local database, WAL mode
-  **Vanilla JS single-page app**
Strict CSP, responsive UI
-  **IMAP4rev1 + IDLE over TLS**
Own client, headers only
-  **scrypt + AES-256-GCM**
Encrypted local vault
-  **RDAP · HIBP · Disconnect · PSL**
Live public data sources
-  **PDF 1.7 writer (own)**
Compliance-grade reports
-  **Any laptop · Win / macOS / Linux**
Runs on 127.0.0.1, Rs 0 cost

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype: live scan of dominos.co.in on the team laptop (score 44/100)



Verified end-to-end: real IMAP sync, leak attribution, quarantine (MOVE) and real-time IDLE alert tested; four compliance PDF reports generated from real records.

FEASIBILITY AND VIABILITY



24 / 24

web-attack security checks passed
(OWASP-based probe)



~2 s

from a leaked email arriving to the
alert (IMAP IDLE)



1–2 s

per live site scan on the team laptop
(3 real sites)

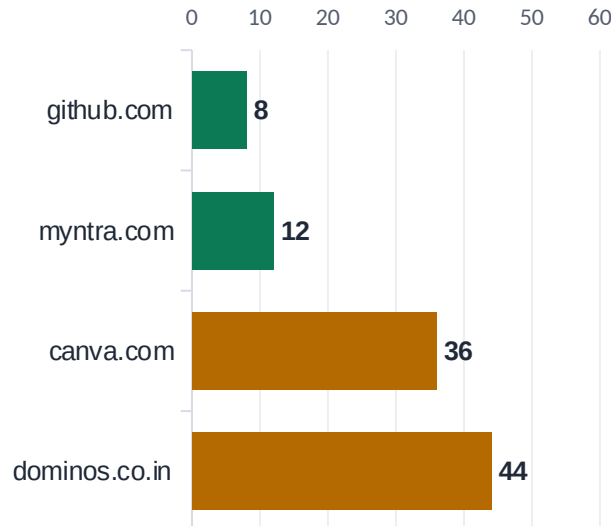


0

third-party packages · Rs 0 running
cost · no cloud server

❖ Analysis of the feasibility of the idea

Live risk score (0–100), real scans



- **Already built and running** — about 4,400 lines in 36 files, installed on the team laptop.
- **Uses free, public data** — RDAP, Have I Been Pwned, Disconnect list, DNS and TLS.
- **Works with any IMAP mailbox** — Gmail, Zoho or your own domain; no API approval needed.
- **No hosting cost** — everything runs locally on 127.0.0.1.
- **Honest scoring** — checks a site refuses are marked unverified and add risk.

❖ Potential challenges and risks → Strategies for overcoming these challenges

Challenge / risk	Strategy
Spammers can strip the +tag from Gmail aliases	Catch-all domain mode gives fully separate addresses
Sites block automated scans (HTTP 403)	Signal marked "unverified" and scored as risk, never as safe
Companies mail from other domains (false leaks)	"Lookalike" review + one-click "same company" trust list
Gmail needs an app password	Guided onboarding, TLS-only, password encrypted with AES-256
DPDP duties on companies start 13 May 2027	Letters cite the Act + own policy today; GDPR where applicable
Local app could itself be attacked	SSRF guard, CSRF/DNS-rebinding checks, 24-check security probe

IMPACT AND BENEFITS

❖ Potential impact on the target audience



Students & job seekers

Sign up to dozens of portals; keep the inbox clean



Online shoppers

Stop sales spam and phishing from sold data



Professionals

Keep work identity out of breach lists



Communities & SMEs

Evidence to hold vendors accountable

❖ Benefits of the solution (social, economic, environmental, etc.)



Social

- Everyday users regain control of who holds their identity
- Evidence lets people exercise DPDP rights, not just complain



Economic

- Free, open and local: no subscription, no cloud bill
- Less phishing and fraud exposure from leaked addresses



Environmental

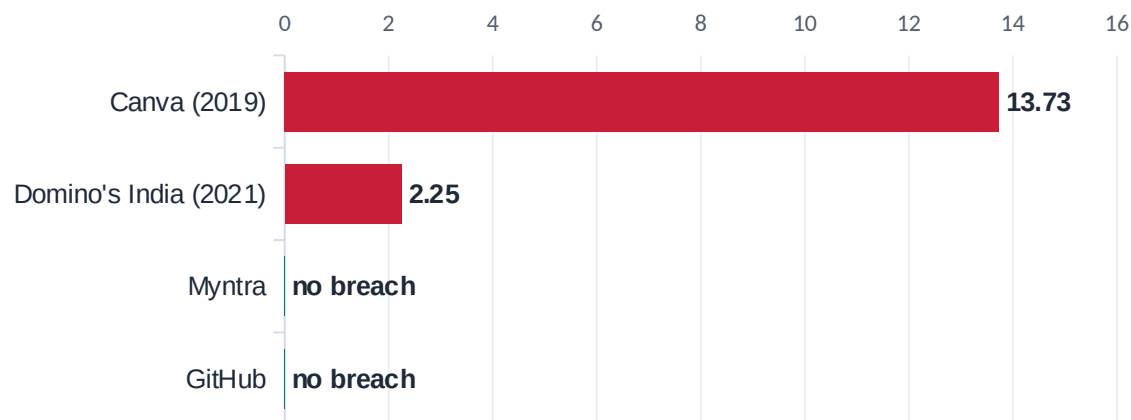
- Runs on the laptop you already own; no servers or data centre
- Reads headers only: minimal data processed and stored



Legal & regulatory

- Supports DPDP s.8(6), 11, 12, 13 and GDPR Art. 17
- Pushes companies towards accountable data handling

Breached accounts (crore), source: HIBP



Without TraceMask

One address shared everywhere, no idea who leaked it

With TraceMask

One alias per site, leaker named in seconds, with proof

RESEARCH AND REFERENCES

❖ Details / Links of the reference and research work

#	Reference	Link
1	Problem Statement P16 dossier (SPEC-P16-187E4A5F4)	Kalpvrksh 2.0 organisers
2	Digital Personal Data Protection Act, 2023	meity.gov.in (official PDF)
3	DPDP Rules, 2025 notified (13 Nov 2025)	static.pib.gov.in
4	DPDP phased implementation timeline	sansalegal.com
5	GDPR Article 17: right to erasure	gdpr-info.eu/art-17-gdpr
6	RFC 3501 IMAP4rev1 / RFC 2177 IMAP IDLE	rfc-editor.org/rfc/rfc3501
7	RFC 8601 Authentication-Results (SPF, DKIM, DMARC)	rfc-editor.org/rfc/rfc8601
8	RDAP domain registration data (RFC 9083)	rdap.org
9	Have I Been Pwned API v3: breach catalogue	haveibeenpwned.com/API/v3
10	Disconnect tracking-protection list	github.com/disconnectme
11	OWASP Top 10 (2021) web application risks	owasp.org/Top10
12	Email alias services compared (Relay, SimpleLogin, DDG...)	github.com/fynks

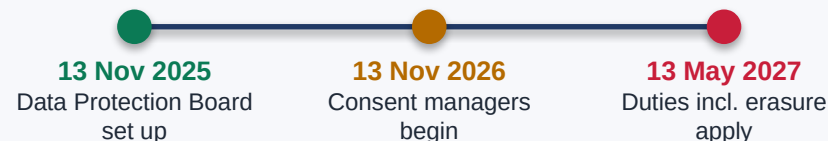
❖ Key research findings



The gap we fill

Alias services exist, but none detects breaches automatically; users must map aliases to sites by hand (ref. 12). TraceMask automates that attribution.

DPDP roll-out (refs. 2–4): why now



Prototype validation (11 Sep 2026)

- **Real scans** of canva.com, dominos.co.in, myntra.com and github.com matched HIBP and RDAP records
- **24/24** OWASP-based security checks passed after fixing 6 findings
- **~2 s** from a leaked email arriving to the leak alert (IMAP IDLE)